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System for off-network access to resources using a remote client

Abstract

Systems, computer program products, and methods are described herein for off-network access to resources using a remote client. The present invention is configured to receive, from one or more client devices, an off-network resource access request from a user; establish one or more network sessions with the one or more client devices; determine, via the one or more network sessions, one or more device identifiers associated with the one or more client devices; retrieve, from an internal repository, a distributed ledger credential associated with the distributed ledger stored on the one or more client devices; unlock, via the one or more network sessions, access to the distributed ledger for the user via the one or more client devices using the distributed ledger credential; and terminate the one or more network sessions in response to unlocking access to the distributed ledger for the user.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention embraces a system for off-network access to resources using a remote client.

BACKGROUND

(2) Typically, to access resources, (e.g., past transactions), each user may be required, by the entity, to authenticate themselves before being given access to the resources. This often requires the user to communicate with the entity, via a network, to present authentication credentials and be subsequently authenticated to access the resources. In some cases, the user may not have access to a network to communicate with the entity, or simply may not wish to connect using a network to communicate with the entity.

(3) Therefore, there is a need for off-network access to resources using a remote client.

SUMMARY

(4) The following presents a simplified summary of one or more embodiments of the present invention, in order to provide a basic understanding of such embodiments. This summary is not an extensive overview of all contemplated embodiments and is intended to neither identify key or critical elements of all embodiments nor delineate the scope of any or all embodiments. Its sole purpose is to present some concepts of one or more embodiments of the present invention in a simplified form as a prelude to the more detailed description that is presented later.

(5) In one aspect, a system for off-network access to resources using a remote client is presented. The system comprising: at least one non-transitory storage device; and at least one processor coupled to the at least one non-transitory storage device, wherein the at least one processor is configured to: receive, from one or more client devices, an off-network resource access request from a user, wherein the one or more client devices comprises a distributed ledger stored thereon; establish one or more network sessions with the one or more client devices; determine, via the one or more network sessions, one or more device identifiers associated with the one or more client devices; retrieve, from an internal repository, a distributed ledger credential associated with the distributed ledger stored on the one or more client devices; unlock, via the one or more network sessions, access to the distributed ledger for the user via the one or more client devices using the distributed ledger credential; and terminate the one or more network sessions in response to unlocking access to the distributed ledger for the user.

(6) In some embodiments, the at least one processor is further configured to: capture information associated with one or more transactions executed by the user using the one or more client devices;

and record the information associated with the one or more transactions in the distributed ledger.

(7) In some embodiments, recording further comprises generating one or more transaction objects for the one or more transactions, wherein the one or more transaction objects comprises at least information associated with the one or more transactions; and deploying the one or more transaction objects on the distributed ledgers, wherein deploying further comprises capturing one or more distributed ledger addresses associated with the recording.

(8) In some embodiments, the at least one processor is further configured to: broadcast the recording to the one or more client devices, wherein broadcasting further comprises updating the one or more distributed ledgers stored on the one or more client devices with the one or more transaction objects.

(9) In some embodiments, the off-network resource access request comprises at least a request to access the one or more transactions stored in the distributed ledger.

(10) In some embodiments, unlocking the access to the distributed ledger further comprises triggering a display of the one or more transactions stored in the distributed ledger for the user via the one or more client devices.

(11) In some embodiments, the at least one processor is further configured to: receive, from the one or more client devices, an escalated access request from the user, wherein the escalated access request is associated with the one or more transactions stored in the distributed ledger; prompt, via the one or more client devices, the user for one or more authentication credentials in response to receiving the escalated access request; receive, via the one or more client devices, the one or more authentication credentials; validate the one or more authentication credentials; and authorize the escalated access request in response to validating the one or more authentication credentials.

(12) In some embodiments, the at least one processor is further configured to: determine that the one or more client devices are programmed to require user authorization prior to allowing the user to communicate the off-network resource access request; and determine that the user is authorized by the one or more client devices to communicate the off-network resource access request based on at least receiving the off-network resource access request from the one or more client devices.

(13) In another aspect, a computer program product for off-network access to resources using a remote client is presented. The computer program product comprising a non-transitory computer-readable medium comprising code causing a first apparatus to: receive, from one or more client devices, an off-network resource access request from a user, wherein the one or more client devices comprises a distributed ledger stored thereon; establish one or more network sessions with the one or more client devices; determine, via the one or more network sessions, one or more device identifiers associated with the one or more client devices; retrieve, from an internal repository, a distributed ledger credential associated with the distributed ledger stored on the one or more client devices; unlock, via the one or more network sessions, access to the distributed ledger for the user via the one or more client devices using the distributed ledger credential; and terminate the one or more network sessions in response to unlocking access to the distributed ledger for the user.

(14) In yet another aspect, a method for off-network access to resources using a remote client is presented. The method comprising: receiving, from one or more client devices, an off-network resource access request from a user, wherein the one or more client devices comprises a distributed ledger stored thereon; establishing one or more network sessions with the one or more client devices; determining, via the one or more network sessions, one or more device identifiers associated with the one or more client devices; retrieving, from an internal repository, a distributed ledger credential associated with the distributed ledger stored on the one or more client devices; unlocking, via the one or more network sessions, access to the distributed ledger for the user via the one or more client devices using the distributed ledger credential; and terminating the one or more network sessions in response to unlocking access to the distributed ledger for the user.

(15) The features, functions, and advantages that have been discussed may be achieved independently in various embodiments of the present invention or may be combined with yet other

embodiments, further details of which can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Having thus described embodiments of the invention in general terms, reference will now be made to the accompanying drawings, wherein:
- (2) FIGS. 1A-1C illustrate technical components of an exemplary distributed computing environment for off-network access to resources using a remote client, in accordance with an embodiment of the invention;
- (3) FIGS. 2A-2B illustrate an exemplary distributed ledger technology (DLT) architecture, in accordance with an embodiment of the invention; and
- (4) FIG. 3 illustrates a process flow for off-network access to resources using a remote client, in accordance with an embodiment of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

- (5) Embodiments of the present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all, embodiments of the invention are shown. Indeed, the invention may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Where possible, any terms expressed in the singular form herein are meant to also include the plural form and vice versa, unless explicitly stated otherwise. Also, as used herein, the term “a” and/or “an” shall mean “one or more,” even though the phrase “one or more” is also used herein. Furthermore, when it is said herein that something is “based on” something else, it may be based on one or more other things as well. In other words, unless expressly indicated otherwise, as used herein “based on” means “based at least in part on” or “based at least partially on.” Like numbers refer to like elements throughout.
- (6) As used herein, an “entity” may be any institution employing information technology resources and particularly technology infrastructure configured for processing large amounts of data. Typically, these data can be related to the people who work for the organization, its products or services, the customers or any other aspect of the operations of the organization. As such, the entity may be any institution, group, association, financial institution, establishment, company, union, authority or the like, employing information technology resources for processing large amounts of data.
- (7) As described herein, a “user” may be an individual associated with an entity. As such, in some embodiments, the user may be an individual having past relationships, current relationships or potential future relationships with an entity. In some embodiments, the user may be an employee (e.g., an associate, a project manager, an IT specialist, a manager, an administrator, an internal operations analyst, or the like) of the entity or enterprises affiliated with the entity.
- (8) As used herein, a “user interface” may be a point of human-computer interaction and communication in a device that allows a user to input information, such as commands or data, into a device, or that allows the device to output information to the user. For example, the user interface includes a graphical user interface (GUI) or an interface to input computer-executable instructions that direct a processor to carry out specific functions. The user interface typically employs certain input and output devices such as a display, mouse, keyboard, button, touchpad, touch screen, microphone, speaker, LED, light, joystick, switch, buzzer, bell, and/or other user input/output device for communicating with one or more users.
- (9) As used herein, an “engine” may refer to core elements of an application, or part of an application that serves as a foundation for a larger piece of software and drives the functionality of

the software. In some embodiments, an engine may be self-contained, but externally-controllable code that encapsulates powerful logic designed to perform or execute a specific type of function. In one aspect, an engine may be underlying source code that establishes file hierarchy, input and output methods, and how a specific part of an application interacts or communicates with other software and/or hardware. The specific components of an engine may vary based on the needs of the specific application as part of the larger piece of software. In some embodiments, an engine may be configured to retrieve resources created in other applications, which may then be ported into the engine for use during specific operational aspects of the engine. An engine may be configurable to be implemented within any general purpose computing system. In doing so, the engine may be configured to execute source code embedded therein to control specific features of the general purpose computing system to execute specific computing operations, thereby transforming the general purpose system into a specific purpose computing system.

(10) As used herein, “authentication credentials” may be any information that can be used to identify of a user. For example, a system may prompt a user to enter authentication information such as a username, a password, a personal identification number (PIN), a passcode, biometric information (e.g., iris recognition, retina scans, fingerprints, finger veins, palm veins, palm prints, digital bone anatomy/structure and positioning (distal phalanges, intermediate phalanges, proximal phalanges, and the like), an answer to a security question, a unique intrinsic user activity, such as making a predefined motion with a user device. This authentication information may be used to authenticate the identity of the user (e.g., determine that the authentication information is associated with the account) and determine that the user has authority to access an account or system. In some embodiments, the system may be owned or operated by an entity. In such embodiments, the entity may employ additional computer systems, such as authentication servers, to validate and certify resources inputted by the plurality of users within the system. The system may further use its authentication servers to certify the identity of users of the system, such that other users may verify the identity of the certified users. In some embodiments, the entity may certify the identity of the users. Furthermore, authentication information or permission may be assigned to or required from a user, application, computing node, computing cluster, or the like to access stored data within at least a portion of the system.

(11) It should also be understood that “operatively coupled,” as used herein, means that the components may be formed integrally with each other, or may be formed separately and coupled together. Furthermore, “operatively coupled” means that the components may be formed directly to each other, or to each other with one or more components located between the components that are operatively coupled together. Furthermore, “operatively coupled” may mean that the components are detachable from each other, or that they are permanently coupled together. Furthermore, operatively coupled components may mean that the components retain at least some freedom of movement in one or more directions or may be rotated about an axis (i.e., rotationally coupled, pivotally coupled). Furthermore, “operatively coupled” may mean that components may be electronically connected and/or in fluid communication with one another.

(12) As used herein, an “interaction” may refer to any communication between one or more users, one or more entities or institutions, one or more devices, nodes, clusters, or systems within the distributed computing environment described herein. For example, an interaction may refer to a transfer of data between devices, an accessing of stored data by one or more nodes of a computing cluster, a transmission of a requested task, or the like.

(13) As used herein, “determining” may encompass a variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, ascertaining, and/or the like. Furthermore, “determining” may also include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory), and/or the like. Also, “determining” may include resolving, selecting, choosing, calculating, establishing, and/or the like. Determining may also include ascertaining that a parameter matches a predetermined criterion,

including that a threshold has been met, passed, exceeded, and so on.

(14) As used herein, a “resource” may generally refer to objects, products, devices, goods, commodities, services, and the like, and/or the ability and opportunity to access and use the same. Some example implementations herein contemplate property held by a user, including property that is stored and/or maintained by a third-party entity. In some example implementations, a resource may be associated with one or more accounts or may be property that is not associated with a specific account.

(15) To access resources, (e.g., past transactions), each user may be required, by the entity, to authenticate themselves before being given access to the resources. This often requires the user to communicate with the entity, via a network, to present authentication credentials and be subsequently authenticated to access the resources. In some cases, the user may not have access to a network to communicate with the entity, or simply may not wish to connect using a network to communicate with the entity. Accordingly, the present invention, (i) Receives, from a client device, an off-network resource access request from a user. The client device may be part of the peer-to-peer DLT network, where the clients are nodes, and each node may include a copy of a distributed ledger stored thereon. The distributed ledger may be associated with the user and may include every transaction executed by the user with the entity. Each time the user executes a transaction using a client device, system may be configured to capture information associated with the transaction and record this information in the distributed ledger. As the distributed ledger is stored locally on the client device, the user may not need to access the network to access these transactions, (ii) Determines a device identifier associated with the client device. A device identifier may be a unique, anonymized string of numbers and letters that identifies a client device. Any time the user communicates an off-network resource access, it is an indication that the user has indeed been verified by the client device locally and is authorized by the client device to communicate the off-network resource access request, (iii) Retrieves, from an internal repository, a distributed ledger credential associated with the distributed ledger stored on the client device, (iv) Unlocks access to the distributed ledger for the user via the client device using the distributed ledger credential. Triggers a display of the one or more transactions stored in the distributed ledger for the user via the client device, (v) more transactions stored in the distributed ledger. The escalated access request may include any additional actions the user wishes to perform in addition to what is currently available for them, i.e., display of the various transactions stored in the distributed ledger. Editing, modifying, and/or rewriting specific aspects of the transaction, attempting to execute an additional transaction, and/or the like, (vi) Prompts, via the client device, the user for one or more authentication credentials, and (vii) On successful validation of the authentication credentials, authorizes the escalated access request.

(16) FIGS. 1A-1C illustrate technical components of an exemplary distributed computing environment for off-network access to resources using a remote client **100**, in accordance with an embodiment of the invention. As shown in FIG. 1A, the distributed computing environment **100** contemplated herein may include a system **130**, a client device(s) **140**, and a network **110** over which the system **130** and client device(s) **140** communicate therebetween. FIG. 1A illustrates only one example of an embodiment of the distributed computing environment **100**, and it will be appreciated that in other embodiments one or more of the systems, devices, and/or servers may be combined into a single system, device, or server, or be made up of multiple systems, devices, or servers. Also, the distributed computing environment **100** may include multiple systems, same or similar to system **130**, with each system providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

(17) In some embodiments, the system **130** and the client device(s) **140** may have a client-server relationship in which the client device(s) **140** are remote devices that request and receive service from a centralized server, i.e., the system **130**. In some other embodiments, the system **130** and the client device(s) **140** may have a peer-to-peer relationship in which the system **140** and the client

device(s) **140** are considered equal and all have the same abilities to use the resources available on the network **110**. Instead of having a central server (e.g., system **130**) which would act as the shared drive, each device that is connect to the network **110** would act as the server for the files stored on it.

(18) The system **130** may represent various forms of servers, such as web servers, database servers, file server, or the like, various forms of digital computing devices, such as laptops, desktops, video recorders, audio/video players, radios, workstations, or the like, or any other auxiliary network devices, such as wearable devices, Internet-of-things devices, electronic kiosk devices, mainframes, or the like, or any combination of the aforementioned.

(19) The client device(s) **140** may represent various forms of electronic devices, including user input devices such as personal digital assistants, cellular telephones, smartphones, laptops, desktops, and other similar computing devices.

(20) The network **110** may be a distributed network that is spread over different networks. This provides a single data communication network, which can be managed jointly or separately by each network. Besides shared communication within the network, the distributed network often also supports distributed processing. The network **110** may be a form of digital communication network such as a telecommunication network, a local area network (“LAN”), a wide area network (“WAN”), a global area network (“GAN”), the Internet, or any combination of the foregoing. The network **110** may be secure and/or unsecure and may also include wireless and/or wired and/or optical interconnection technology.

(21) It is to be understood that the structure of the distributed computing environment and its components, connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed in this document. In one example, the distributed computing environment **100** may include more, fewer, or different components. In another example, some or all of the portions of the distributed computing environment **100** may be combined into a single portion or all of the portions of the system **130** may be separated into two or more distinct portions.

(22) FIG. **1B** illustrates an exemplary component-level structure of the system **130**, in accordance with an embodiment of the invention. As shown in FIG. **1B**, the system **130** may include a processor **102**, memory **104**, input/output (I/O) device **116**, and a storage device **110**. The system **130** may also include a high-speed interface **108** connecting to the memory **104**, and a low-speed interface **112** connecting to low speed bus **114** and storage device **110**. Each of the components **102**, **104**, **108**, **110**, and **112** may be operatively coupled to one another using various buses and may be mounted on a common motherboard or in other manners as appropriate. As described herein, the processor **102** may include a number of subsystems to execute the portions of processes described herein. Each subsystem may be a self-contained component of a larger system (e.g., system **130**) and capable of being configured to execute specialized processes as part of the larger system.

(23) The processor **102** can process instructions, such as instructions of an application that may perform the functions disclosed herein. These instructions may be stored in the memory **104** or on the storage device **110**, for execution within the system **130** using any subsystems described herein. It is to be understood that the system **130** may use, as appropriate, multiple processors, along with multiple memories, and/or I/O devices, to execute the processes described herein.

(24) The memory **104** stores information within the system **130**. In one implementation, the memory **104** is a volatile memory unit or units, such as volatile random access memory (RAM) having a cache area for the temporary storage of information, such as a command, a current operating state of the distributed computing environment **100**, an intended operating state of the distributed computing environment **100**, instructions related to various methods and/or functionalities described herein, and/or the like. In another implementation, the memory **104** is a non-volatile memory unit or units. The memory **104** may also be another form of computer-

readable medium, such as a magnetic or optical disk, which may be embedded and/or may be removable. The non-volatile memory may additionally or alternatively include an EEPROM, flash memory, and/or the like for storage of information such as instructions and/or data that may be read during execution of computer instructions. The memory **104** may store, recall, receive, transmit, and/or access various files and/or information used by the system **130** during operation.

(25) The storage device **106** is capable of providing mass storage for the system **130**. In one aspect, the storage device **106** may be or contain a computer-readable medium, such as a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. A computer program product can be tangibly embodied in an information carrier. The computer program product may also contain instructions that, when executed, perform one or more methods, such as those described above. The information carrier may be a non-transitory computer- or machine-readable storage medium, such as the memory **104**, the storage device **104**, or memory on processor **102**.

(26) The high-speed interface **108** manages bandwidth-intensive operations for the system **130**, while the low speed controller **112** manages lower bandwidth-intensive operations. Such allocation of functions is exemplary only. In some embodiments, the high-speed interface **108** is coupled to memory **104**, input/output (I/O) device **116** (e.g., through a graphics processor or accelerator), and to high-speed expansion ports **111**, which may accept various expansion cards (not shown). In such an implementation, low-speed controller **112** is coupled to storage device **106** and low-speed expansion port **114**. The low-speed expansion port **114**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g., through a network adapter.

(27) The system **130** may be implemented in a number of different forms. For example, it may be implemented as a standard server, or multiple times in a group of such servers. Additionally, the system **130** may also be implemented as part of a rack server system or a personal computer such as a laptop computer. Alternatively, components from system **130** may be combined with one or more other same or similar systems and an entire system **130** may be made up of multiple computing devices communicating with each other.

(28) FIG. 1C illustrates an exemplary component-level structure of the client device(s) **140**, in accordance with an embodiment of the invention. As shown in FIG. 1C, the client device(s) **140** includes a processor **152**, memory **154**, an input/output device such as a display **156**, a communication interface **158**, and a transceiver **160**, among other components. The client device(s) **140** may also be provided with a storage device, such as a microdrive or other device, to provide additional storage. Each of the components **152**, **154**, **158**, and **160**, are interconnected using various buses, and several of the components may be mounted on a common motherboard or in other manners as appropriate.

(29) The processor **152** is configured to execute instructions within the client device(s) **140**, including instructions stored in the memory **154**, which in one embodiment includes the instructions of an application that may perform the functions disclosed herein, including certain logic, data processing, and data storing functions. The processor may be implemented as a chipset of chips that include separate and multiple analog and digital processors. The processor may be configured to provide, for example, for coordination of the other components of the client device(s) **140**, such as control of user interfaces, applications run by client device(s) **140**, and wireless communication by client device(s) **140**.

(30) The processor **152** may be configured to communicate with the user through control interface **164** and display interface **166** coupled to a display **156**. The display **156** may be, for example, a TFT LCD (Thin-Film-Transistor Liquid Crystal Display) or an OLED (Organic Light Emitting Diode) display, or other appropriate display technology. The display interface **156** may comprise

appropriate circuitry and configured for driving the display **156** to present graphical and other information to a user. The control interface **164** may receive commands from a user and convert them for submission to the processor **152**. In addition, an external interface **168** may be provided in communication with processor **152**, so as to enable near area communication of client device(s) **140** with other devices. External interface **168** may provide, for example, for wired communication in some implementations, or for wireless communication in other implementations, and multiple interfaces may also be used.

(31) The memory **154** stores information within the client device(s) **140**. The memory **154** can be implemented as one or more of a computer-readable medium or media, a volatile memory unit or units, or a non-volatile memory unit or units. Expansion memory may also be provided and connected to client device(s) **140** through an expansion interface (not shown), which may include, for example, a SIMM (Single In Line Memory Module) card interface. Such expansion memory may provide extra storage space for client device(s) **140** or may also store applications or other information therein. In some embodiments, expansion memory may include instructions to carry out or supplement the processes described above and may include secure information also. For example, expansion memory may be provided as a security module for client device(s) **140** and may be programmed with instructions that permit secure use of client device(s) **140**. In addition, secure applications may be provided via the SIMM cards, along with additional information, such as placing identifying information on the SIMM card in a non-hackable manner.

(32) The memory **154** may include, for example, flash memory and/or NVRAM memory. In one aspect, a computer program product is tangibly embodied in an information carrier. The computer program product contains instructions that, when executed, perform one or more methods, such as those described herein. The information carrier is a computer-or machine-readable medium, such as the memory **154**, expansion memory, memory on processor **152**, or a propagated signal that may be received, for example, over transceiver **160** or external interface **168**.

(33) In some embodiments, the user may use the client device(s) **140** to transmit and/or receive information or commands to and from the system **130** via the network **110**. Any communication between the system **130** and the client device(s) **140** may be subject to an authentication protocol allowing the system **130** to maintain security by permitting only authenticated users (or processes) to access the protected resources of the system **130**, which may include servers, databases, applications, and/or any of the components described herein. To this end, the system **130** may trigger an authentication subsystem that may require the user (or process) to provide authentication credentials to determine whether the user (or process) is eligible to access the protected resources. Once the authentication credentials are validated and the user (or process) is authenticated, the authentication subsystem may provide the user (or process) with permissioned access to the protected resources. Similarly, the client device(s) **140** may provide the system **130** (or other client devices) permissioned access to the protected resources of the client device(s) **140**, which may include a GPS device, an image capturing component (e.g., camera), a microphone, and/or a speaker.

(34) The client device(s) **140** may communicate with the system **130** through communication interface **158**, which may include digital signal processing circuitry where necessary. Communication interface **158** may provide for communications under various modes or protocols, such as the Internet Protocol (IP) suite (commonly known as TCP/IP). Protocols in the IP suite define end-to-end data handling methods for everything from packetizing, addressing and routing, to receiving. Broken down into layers, the IP suite includes the link layer, containing communication methods for data that remains within a single network segment (link); the Internet layer, providing internetworking between independent networks; the transport layer, handling host-to-host communication; and the application layer, providing process-to-process data exchange for applications. Each layer contains a stack of protocols used for communications. In addition, the communication interface **158** may provide for communications under various telecommunications

standards (2G, 3G, 4G, 5G, and/or the like) using their respective layered protocol stacks. These communications may occur through a transceiver **160**, such as radio-frequency transceiver. In addition, short-range communication may occur, such as using a Bluetooth, Wi-Fi, or other such transceiver (not shown). In addition, GPS (Global Positioning System) receiver module **170** may provide additional navigation- and location-related wireless data to client device(s) **140**, which may be used as appropriate by applications running thereon, and in some embodiments, one or more applications operating on the system **130**.

(35) The client device(s) **140** may also communicate audibly using audio codec **162**, which may receive spoken information from a user and convert it to usable digital information. Audio codec **162** may likewise generate audible sound for a user, such as through a speaker, e.g., in a handset of client device(s) **140**. Such sound may include sound from voice telephone calls, may include recorded sound (e.g., voice messages, music files, etc.) and may also include sound generated by one or more applications operating on the client device(s) **140**, and in some embodiments, one or more applications operating on the system **130**.

(36) Various implementations of the distributed computing environment **100**, including the system **130** and client device(s) **140**, and techniques described here can be realized in digital electronic circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof.

(37) FIGS. 2A-2B illustrate an exemplary distributed ledger technology (DLT) architecture, in accordance with an embodiment of the invention. DLT may refer to the protocols and supporting infrastructure that allow computing devices (peers) in different locations to propose and validate transactions and update records in a synchronized way across a network. Accordingly, DLT is based on a decentralized model, in which these peers collaborate and build trust over the network. To this end, DLT involves the use of potentially peer-to-peer protocol for a cryptographically secured distributed ledger of transactions represented as transaction objects that are linked. As transaction objects each contain information about the transaction object previous to it, they are linked with each additional transaction object, reinforcing the ones before it. Therefore, distributed ledgers are resistant to modification of their data because once recorded, the data in any given transaction object cannot be altered retroactively without altering all subsequent transaction objects.

(38) To permit transactions and agreements to be carried out among various peers without the need for a central authority or external enforcement mechanism, DLT uses smart contracts. Smart contracts are computer code that automatically executes all or parts of an agreement and is stored on a DLT platform. The code can either be the sole manifestation of the agreement between the parties or might complement a traditional text-based contract and execute certain provisions, such as transferring funds from Party A to Party B. The code itself is replicated across multiple nodes (peers) and, therefore, benefits from the security, permanence, and immutability that a distributed ledger offers. That replication also means that as each new transaction object is added to the distributed ledger, the code is, in effect, executed. If the parties have indicated, by initiating a transaction, that certain parameters have been met, the code will execute the step triggered by those parameters. If no such transaction has been initiated, the code will not take any steps.

(39) Various other specific-purpose implementations of distributed ledgers have been developed. These include distributed domain name management, decentralized crowd-funding, synchronous/asynchronous communication, decentralized real-time ride sharing and even a general purpose deployment of decentralized applications. In some embodiments, a distributed ledger may be characterized as a public distributed ledger, a consortium distributed ledger, or a private distributed ledger. A public distributed ledger is a distributed ledger that anyone in the world can read, anyone in the world can send transactions to and expect to see them included if they are valid, and anyone in the world can participate in the consensus process for determining which transaction objects get added to the distributed ledger and what the current state each transaction object is. A public block chain is generally considered to be fully decentralized. On the other hand, fully private

distributed ledger is a distributed ledger whereby permissions are kept centralized with one entity. The permissions may be public or restricted to an arbitrary extent. And lastly, a consortium distributed ledger is a distributed ledger where the consensus process is controlled by a pre-selected set of nodes; for example, a distributed ledger may be associated with a number of member institutions (say 15), each of which operate in such a way that the at least 10 members must sign every transaction object in order for the transaction object to be valid. The right to read such a distributed ledger may be public or restricted to the participants. These distributed ledgers may be considered partially decentralized.

(40) As shown in FIG. 2A, the exemplary DLT architecture **200** includes a distributed ledger **204** being maintained on multiple devices (nodes) **202** that are authorized to keep track of the distributed ledger **204**. For example, these nodes **202** may be computing devices such as system **130** and client device(s) **130**. One node **202** in the DLT architecture **200** may have a complete or partial copy of the entire distributed ledger **204** or set of transactions and/or transaction objects **204A** on the distributed ledger **204**. Transactions are initiated at a node and communicated to the various nodes of the block chain. Any of the nodes can validate a transaction, record the transaction to its copy of the distributed ledger, and/or broadcast the transaction, its validation (in the form of a transaction object) and/or other data to other nodes.

(41) As shown in FIG. 2B, an exemplary transaction object **204A** may include a transaction header **206** and a transaction object data **208**. The transaction header **206** may include a cryptographic hash of the previous transaction object **206A**, a nonce **206B**—a randomly generated 32-bit whole number when the transaction object is created, cryptographic hash of the current transaction object **206C** wedded to the nonce **206B**, and a time stamp **206D**. The transaction object data **208** may include transaction information **208A** being recorded. Once the transaction object **204A** is generated, the transaction information **208A** is considered signed and forever tied to its nonce **206B** and hash **206C**. Once generated, the transaction object **204A** is then recorded in the distributed ledger **204**. deployed on the distributed ledger **204**. At this time, a distributed ledger address is generated for the transaction object **204A**, i.e., an indication of where it is located on the distributed ledger **204** and captured for recording purposes. Once deployed, the transaction information **208A** is considered recorded in the distributed ledger **204**.

(42) FIG. 3 illustrates a process flow for off-network access to resources using a remote client, in accordance with an embodiment of the invention. As shown in block **302**, the process flow includes receiving, from one or more client devices, an off-network resource access request from a user. In some embodiments, wherein each client device may be part of the peer-to-peer DLT network, where the clients are nodes, and each node may include a copy of a distributed ledger stored thereon. Here, the distributed ledger may be associated with the user and may include every transaction executed by the user with the entity.

(43) Each time the user executes a transaction using a client device, system may be configured to capture information associated with the transaction and record this information in the distributed ledger. If no distributed ledger exists for the user, the system may be configured to create a distributed ledger to record the current transaction and maintain an immutable record of the current and any future transaction executed by the user with the entity. As the transaction information may include sensitive data, the distributed ledger used herein may be a private distributed ledger where access privileges and protected and are kept centralized with the entity.

(44) As described herein, to record the transaction information in the distributed ledger, the system may be configured to generate transaction object for the transaction. Here, the transaction object may store, among other items, the transaction information. This transaction object is then deployed on the distributed ledger. Once deployed the system may be configured to broadcast the recording to the one or more client devices, thereby updating the copies of the distributed ledgers stored on each of the client devices with the newly recorded transaction object. In some embodiments, broadcasting the recording may be done in the form of a silent notification, i.e., a notification with

no audible or visual alert, providing distributed systems to be notified with the transaction object to update distributed ledger stored on the client device.

(45) Typically, to access resources, (e.g., past transactions), each user may be required, by the entity, to authenticate themselves before being given access to the resources. This often requires the user to communicate with the entity, via a network, to present authentication credentials and be subsequently authenticated to access the resources. In some cases, the user may not have access to a network to communicate with the entity, or simply may not wish to connect using a network to communicate with the entity. In these situations, as these distributed ledgers are stored locally on the client devices, the user may not need to access the network to communicate with the entity. By requesting an off-network resource access request, the user may access the resources stored in the distributed ledger.

(46) To enable such access, the system may be configured to require that the client device be programmed to validate authentication credentials of the user locally before communicating the off-network resource request to the entity. Therefore, any time the user communicates an off-network resource access, it is an indication that the user has indeed been verified by the client device locally, and is authorized by the client device to communicate the off-network resource access request.

(47) Next, as shown in block **304**, the process flow includes establishing one or more network sessions with the one or more client devices. As described herein, a network session may be a temporary and interactive information interchange between two or more communicating devices, or between an entity and user. Next, as shown in block **306**, the process flow includes determining, via the one or more network sessions, one or more device identifiers associated with the one or more client devices. In some embodiments, a device identifier may be a unique, anonymized string of numbers and letters that identifies a client device. It is stored on the client device and made retrievable in specific situations. Here, the system may be configured to retrieve the device identifier in response to receiving the off-network resource access request.

(48) Next, as shown in block **308**, the process flow includes retrieving, from an internal repository, a distributed ledger credential associated with the distributed ledger stored on the one or more client devices. In some embodiments, the distributed ledger credential for the distributed ledger may be the same for each copy of the distributed ledger. In some other embodiments, the distributed ledger credential for the distributed ledger may depend on the specific copy of the distributed ledger and/or the client device on which that particular copy of the distributed ledger is being stored. By identifying, using the device identifier, the particular client device that is requesting the off-network resource access, the system may be configured to retrieve a distributed ledger credential that is specific to the copy that is stored on that particular client device.

(49) Next, as shown in block **310**, the process flow includes unlocking, via the one or more network sessions, access to the distributed ledger for the user via the one or more client devices using the distributed ledger credential. In some embodiments, unlocking the access to the distributed ledger may include triggering a display of the one or more transactions stored in the distributed ledger for the user via the one or more client devices.

(50) Next, as shown in block **312**, the process flow includes terminating the one or more network sessions in response to unlocking access to the distributed ledger for the user. In some embodiments, once the distributed ledger is unlocked, then the network session may immediately terminate, allowing the user to truly access the distributed ledger off-network. In some embodiments, the system may be configured to receive, from the one or more client devices, an escalated access request from the user for the one or more transactions stored in the distributed ledger. In one aspect, the escalated access request may include any additional actions the user wishes to perform in addition to what is currently available for them, i.e., display of the various transactions stored in the distributed ledger. These additional actions may include editing, modifying, and/or rewriting specific aspects of the transaction, attempting to execute an additional

transaction, and/or the like. In response, the system may be configured to prompt, via the one or more client devices, the user for one or more authentication credentials. In response, the system may be configured to receive, via the one or more client devices, the one or more authentication credentials for validation. On successful validation, the system may be configured to authorize the escalated access request.

(51) As will be appreciated by one of ordinary skill in the art, the present invention may be embodied as an apparatus (including, for example, a system, a machine, a device, a computer program product, and/or the like), as a method (including, for example, a business process, a computer-implemented process, and/or the like), or as any combination of the foregoing.

Accordingly, embodiments of the present invention may take the form of an entirely software embodiment (including firmware, resident software, micro-code, and the like), an entirely hardware embodiment, or an embodiment combining software and hardware aspects that may generally be referred to herein as a “system.” Furthermore, embodiments of the present invention may take the form of a computer program product that includes a computer-readable storage medium having computer-executable program code portions stored therein. As used herein, a processor may be “configured to” perform a certain function in a variety of ways, including, for example, by having one or more special-purpose circuits perform the functions by executing one or more computer-executable program code portions embodied in a computer-readable medium, and/or having one or more application-specific circuits perform the function.

(52) It will be understood that any suitable computer-readable medium may be utilized. The computer-readable medium may include, but is not limited to, a non-transitory computer-readable medium, such as a tangible electronic, magnetic, optical, infrared, electromagnetic, and/or semiconductor system, apparatus, and/or device. For example, in some embodiments, the non-transitory computer-readable medium includes a tangible medium such as a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a compact disc read-only memory (CD-ROM), and/or some other tangible optical and/or magnetic storage device. In other embodiments of the present invention, however, the computer-readable medium may be transitory, such as a propagation signal including computer-executable program code portions embodied therein.

(53) It will also be understood that one or more computer-executable program code portions for carrying out the specialized operations of the present invention may be required on the specialized computer include object-oriented, scripted, and/or unscripted programming languages, such as, for example, Java, Perl, Smalltalk, C++, SAS, SQL, Python, Objective C, and/or the like. In some embodiments, the one or more computer-executable program code portions for carrying out operations of embodiments of the present invention are written in conventional procedural programming languages, such as the “C” programming languages and/or similar programming languages. The computer program code may alternatively or additionally be written in one or more multi-paradigm programming languages, such as, for example, F #.

(54) It will further be understood that some embodiments of the present invention are described herein with reference to flowchart illustrations and/or block diagrams of systems, methods, and/or computer program products. It will be understood that each block included in the flowchart illustrations and/or block diagrams, and combinations of blocks included in the flowchart illustrations and/or block diagrams, may be implemented by one or more computer-executable program code portions. These computer-executable program code portions execute via the processor of the computer and/or other programmable data processing apparatus and create mechanisms for implementing the steps and/or functions represented by the flowchart(s) and/or block diagram block(s).

(55) It will also be understood that the one or more computer-executable program code portions may be stored in a transitory or non-transitory computer-readable medium (e.g., a memory, and the

like) that can direct a computer and/or other programmable data processing apparatus to function in a particular manner, such that the computer-executable program code portions stored in the computer-readable medium produce an article of manufacture, including instruction mechanisms which implement the steps and/or functions specified in the flowchart(s) and/or block diagram block(s).

(56) The one or more computer-executable program code portions may also be loaded onto a computer and/or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer and/or other programmable apparatus. In some embodiments, this produces a computer-implemented process such that the one or more computer-executable program code portions which execute on the computer and/or other programmable apparatus provide operational steps to implement the steps specified in the flowchart(s) and/or the functions specified in the block diagram block(s). Alternatively, computer-implemented steps may be combined with operator and/or human-implemented steps in order to carry out an embodiment of the present invention.

(57) While certain exemplary embodiments have been described and shown in the accompanying drawings, it is to be understood that such embodiments are merely illustrative of, and not restrictive on, the broad invention, and that this invention not be limited to the specific constructions and arrangements shown and described, since various other changes, combinations, omissions, modifications and substitutions, in addition to those set forth in the above paragraphs, are possible. Those skilled in the art will appreciate that various adaptations and modifications of the just described embodiments can be configured without departing from the scope and spirit of the invention. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described herein.

Claims

1. A system for off-network access to resources using a remote client, the system comprising: a processor; a non-transitory storage device containing instructions that, when executed by the processor, cause the processor to: receive, from a client device, an off-network resource access request from a user, wherein the a client device comprises a distributed ledger stored thereon; establish a network session with the a client device; determine, via the one or more network sessions, a device identifier associated with the a client device; retrieve, from an internal repository, a distributed ledger credential associated with a copy of the distributed ledger that is stored on the a client device, wherein the distributed ledger credential is specific to the copy of the distributed ledger and the client device on which it is stored; unlock, via the network session, access to the distributed ledger for the user via the a client device using the distributed ledger credential; terminate the one or more network session in response to unlocking access to the distributed ledger for the user to enable off-network distributed ledger access; receive, from the client device, an escalated access request from the user to perform additional actions to one or more transactions stored in the copy of the distributed ledger, wherein the additional actions are previously unavailable to the user; prompt, via the client device, the user for one or more authentication credentials in response to receiving the escalated access request; receive, via the client device, the one or more authentication credentials; validate the one or more authentication credentials; and authorize the user for escalated access request using the client device in response to validating the one or more authentication credentials.
2. The system of claim 1, wherein the instructions, when executed, further cause the processor to: capture information associated with the one or more transactions executed by the user using the a client device; and record the information associated with the one or more transactions in the distributed ledger.
3. The system of claim 2, wherein recording further comprises: generating one or more transaction

objects for the one or more transactions, wherein the one or more transaction objects comprises at least information associated with the one or more transactions; and deploying the one or more transaction objects on the distributed ledgers, wherein deploying further comprises capturing one or more distributed ledger addresses associated with the recording.

4. The system of claim 3, wherein the instructions, when executed, further cause the processor to: broadcast the recording to a plurality of client devices, wherein broadcasting further comprises updating the one or more distributed ledgers stored on the plurality of client devices with the one or more transaction objects.

5. The system of claim 2, wherein the off-network resource access request comprises at least a request to access the one or more transactions stored in the distributed ledger.

6. The system of claim 5, wherein unlocking the access to the distributed ledger further comprises triggering a display of the one or more transactions stored in the distributed ledger for the user via the a client device.

7. The system of claim 1, wherein the instructions, when executed, further cause the processor to: determine that the a client device is programmed to require user authorization prior to allowing the user to communicate the off-network resource access request; and determine that the user is authorized by the a client device to communicate the off-network resource access request based on at least receiving the off-network resource access request from the a client device.

8. A computer program product for off-network access to resources using a remote client, the computer program product comprising a non-transitory computer-readable medium comprising code causing a first apparatus to: receive, from a client device, an off-network resource access request from a user, wherein the a client device comprises a distributed ledger stored thereon; establish a network session with the a client device; determine, via the one or more network sessions, a device identifier associated with the a client device; retrieve, from an internal repository, a distributed ledger credential associated with a copy of the distributed ledger that is stored on the a client device, wherein the distributed ledger credential is specific to the copy of the distributed ledger and the client device on which it is stored; unlock, via the one or more network session, access to the distributed ledger for the user via the a client device using the distributed ledger credential; terminate the one or more network session in response to unlocking access to the distributed ledger for the user to enable off-network distributed ledger access; receive, from the client device, an escalated access request from the user to perform additional actions to one or more transactions stored in the copy of the distributed ledger, wherein the additional actions are previously unavailable to the user; prompt, via the client device, the user for one or more authentication credentials in response to receiving the escalated access request; receive, via the client device, the one or more authentication credentials; validate the one or more authentication credentials; and authorize the user for escalated access request using the client device in response to validating the one or more authentication credentials.

9. The computer program product of claim 8, wherein the code, when executed, further causes the first apparatus to: capture information associated with the one or more transactions executed by the user using the a client device; and record the information associated with the one or more transactions in the distributed ledger.

10. The computer program product of claim 9, wherein recording further comprises: generating one or more transaction objects for the one or more transactions, wherein the one or more transaction objects comprises at least information associated with the one or more transactions; and deploying the one or more transaction objects on the distributed ledgers, wherein deploying further comprises capturing one or more distributed ledger addresses associated with the recording.

11. The computer program product of claim 10, wherein the code, when executed, further causes the first apparatus to: broadcast the recording to a plurality of client devices, wherein broadcasting further comprises updating the one or more distributed ledgers stored on the plurality of client devices with the one or more transaction objects.

12. The computer program product of claim 9, wherein the off-network resource access request comprises at least a request to access the one or more transactions stored in the distributed ledger.

13. The computer program product of claim 12, wherein unlocking the access to the distributed ledger further comprises triggering a display of the one or more transactions stored in the distributed ledger for the user via the a client device.

14. The computer program product of claim 8, wherein the code, when executed, further causes the first apparatus to: determine that the a client device is programmed to require user authorization prior to allowing the user to communicate the off-network resource access request; and determine that the user is authorized by the a client device to communicate the off-network resource access request based on at least receiving the off-network resource access request from the a client device.

15. A method for off-network access to resources using a remote client, the method comprising: receiving, from a client device, an off-network resource access request from a user, wherein the a client device comprises a distributed ledger stored thereon; establishing a network session with the a client device; determining, via the one or more network sessions, a device identifier associated with the a client device; retrieving, from an internal repository, a distributed ledger credential associated with a copy of the distributed ledger that is stored on the a client device, wherein the distributed ledger credential is specific to the copy of the distributed ledger and the client device on which it is stored; unlocking, via the one or more network session, access to the distributed ledger for the user via the a client device using the distributed ledger credential; terminating the one or more network session in response to unlocking access to the distributed ledger for the user to enable off-network distributed ledger access; receiving, from the client device, an escalated access request from the user to perform additional actions to one or more transactions stored in the copy of the distributed ledger, wherein the additional actions are previously unavailable to the user; prompting, via the client device, the user for one or more authentication credentials in response to receiving the escalated access request; receiving, via the client device, the one or more authentication credentials; validating the one or more authentication credentials; and authorizing the user for escalated access request using the client device in response to validating the one or more authentication credentials.

16. The method of claim 15, wherein the method further comprises: capturing information associated with the one or more transactions executed by the user using the a client device; and recording the information associated with the one or more transactions in the distributed ledger.

17. The method of claim 16, wherein recording further comprises: generating one or more transaction objects for the one or more transactions, wherein the one or more transaction objects comprises at least information associated with the one or more transactions; and deploying the one or more transaction objects on the distributed ledgers, wherein deploying further comprises capturing one or more distributed ledger addresses associated with the recording.

18. The method of claim 17, wherein the method further comprises: broadcast the recording to a plurality of client devices, wherein broadcasting further comprises updating the one or more distributed ledgers stored on the plurality of client devices with the one or more transaction objects.
